

# Practice Quiz 06C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 06

*Existence and uniqueness as two separate claims.*

Label and prove both halves.

1. Prove that for every real number  $a \neq 0$ , there is exactly one real number  $x$  such that  $a(x - 1) = 4$ .

**Solution** Let  $a \in \mathbf{R}$  and suppose  $a \neq 0$ .

*Existence.* Set  $x = 1 + 4/a$ . This is real because  $a \neq 0$ , and

$$a(x - 1) = a \left( 1 + \frac{4}{a} - 1 \right) = 4.$$

Thus a solution exists.

*Uniqueness.* Suppose  $x$  and  $y$  both satisfy the equation. Then  $a(x - 1) = 4 = a(y - 1)$ . Subtracting gives  $a(x - y) = 0$ . Since  $a \neq 0$ , we have  $x = y$ . Thus the solution is unique. ■